CLASS 9 MATHEMATICS • NCERT 0904CH04

Exploring Algebraic Identities

An exhaustive, point-by-point classroom presentation covering geometrical visualizations, factorisation techniques, algebra tiles, trinomial expansions, Śhrīdharāchārya's method, cubic binomials, and rational algebraic simplifications.

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The Consecutive Squares Mystery: The Invariable 2

1 Trial 1: 1, 4, 9

- Smallest: 1, Largest: 9, Middle: 4
- Sum of extremes = 1 + 9 = **10**
- Twice middle = 2 × 4 = **8**
- Difference: 10 − 8 = **2**

2 Trial 2: 9, 16, 25

- Smallest: 9, Largest: 25, Middle: 16
- Sum of extremes = 9 + 25 = **34**
- Twice middle = 2 × 16 = **32**
- Difference: 34 − 32 = **2**

3 Trial 3: 25, 36, 49

- Smallest: 25, Largest: 49, Middle: 36
- Sum of extremes = 25 + 49 = **74**
- Twice middle = 2 × 36 = **72**
- Difference: 74 − 72 = **2**

Why Is The Result Always 2?

This mathematical miracle is not an accident of arithmetic. It is a direct consequence of an underlying algebraic identity that holds true for every integer n across the infinite number line!

Check Your Understanding: Consecutive Squares

Q1 Test the consecutive squares rule on the triple 36, 49, and 64. What is the final result?

Q2 If three consecutive integers are represented algebraically as (n − 1), n, and (n + 1), what are the expressions for their squares?

Q3 Does the consecutive square difference rule depend on whether n is small or astronomically large?

Q4 What pattern emerges if you examine four consecutive squares instead of three?

Visualising $(a + b)^2$: Geometrical Model & Identities

Dissection of Square of Side (a + b)

• Construct a square of side **(a + b) units**.

• Area of total outer square = **$(a + b)^2$**.

• Partitioned into 4 sub-regions:
— Square of area **a^2**
— Square of area **b^2**
— Two congruent rectangles, each of area **ab**

$$(a + b)^2 = a^2 + 2ab + b^2$$

THE FUNDAMENTAL BINOMIAL SQUARE IDENTITY

Equation vs. Algebraic Identity

- **Equation:** True only for *specific* values of the variable.
  Example: $x^2 - 1 = 24$ is true only for $x = 5$ or $x = -5$.
- **Algebraic Identity:** An equality that remains *universally true* for **ALL possible values** of the variables!
  Example: $(x + y)^2 = x^2 + 2xy + y^2$ holds for all real x, y.
- Verified algebraically via Distributivity:
  $(a + b)(a + b) = a(a + b) + b(a + b) = a^2 + 2ab + b^2$.

Critical Reflection: Why $(a + b)^2 \neq a^2 + b^2$

Because $(a + b)^2 - (a^2 + b^2) =$ **$2ab$**. When a and b share the same sign ($ab > 0$), $(a + b)^2$ is strictly greater than $a^2 + b^2$ by exactly twice their product!

Check Your Understanding: $(a + b)^2$ & Identities

Q1 What is the fundamental difference between an algebraic equation and an algebraic identity?

Q2 Under what specific condition for real numbers a and b will $(a + b)^2$ be equal to $a^2 + b^2$?

Q3 Using the identity $(a + b)^2$, how can you mentally compute 43^2 without direct multiplication?

Q4 If $a = -2$ and $b = -3$, does $(a + b)^2$ still equal $a^2 + 2ab + b^2$? Verify mentally.

The Identity (a – b)² & Factorisation Techniques

Derivation & Geometric Model

• Substitute **b** → **–b** in (a + b)²:

$$(a - b)^2 = a^2 - 2ab + b^2$$

SQUARE OF A BINOMIAL DIFFERENCE

• **Geometry (Fig 4.3):** Start with square of side a (area a²). Subtract rectangle ab and rectangle b(a – b).
Area = a² – ab – b(a – b) = **a² – 2ab + b²**.

• *Fast Squaring:* 29² = (30 – 1)² = 900 – 60 + 1 = **841**.

Proof of the Consecutive Squares Mystery

Let three consecutive integers be **(n – 1), n, (n + 1)**.

• Sum of extreme squares:
(n – 1)² + (n + 1)² = (n² – 2n + 1) + (n² + 2n + 1)
= **2n² + 2**

• Subtract twice the middle square (2n²):
(2n² + 2) – 2n² = **2** ✅

Proved algebraically for every possible integer n!

Pro-Tip: Extract Common Monomial Factors First!

To factor 50p² + 60pq + 18q², first pull out the common factor 2: 2(25p² + 30pq + 9q²) = 2[(5p)² + 2(5p)(3q) + (3q)²] = **2(5p + 3q)²**.

Check Your Understanding: $(a - b)^2$ & Common Factors

Q1 How does substituting -b in place of b in $(a + b)^2$ naturally produce the identity for $(a - b)^2$?

Q2 Why did the term $-2n$ and $+2n$ cancel out when proving the consecutive squares rule?

Q3 How would you use $(a - b)^2$ to quickly evaluate 79^2 without long multiplication?

Q4 In factoring $3a^2 + 4ab + (4/3)b^2$, what common factor should be factored out first?

The Square of a Trinomial: $(a + b + c)^2$

Derivation by Substitution

• Treat **(b + c) as a single variable d**.

• $(a + d)^2 = a^2 + 2ad + d^2$

• Replace d with (b + c):
$= a^2 + 2a(b + c) + (b + c)^2$
$= a^2 + 2ab + 2ac + (b^2 + 2bc + c^2)$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

TRINOMIAL SQUARE IDENTITY

Geometrical Dissection (Fig 4.4)

A square of side (a + b + c) is partitioned into **9 regions**:

- **3 Squares:** a^2, b^2, c^2
- **6 Rectangles (3 symmetric pairs):**
  - Two rectangles of area **ab**
  - Two rectangles of area **bc**
  - Two rectangles of area **ca**
- Total Area = $\mathbf{a^2 + b^2 + c^2 + 2ab + 2bc + 2ca}$.

Application: Fast Evaluation of 119^2

$119^2 = (100 + 10 + 9)^2 = 100^2 + 10^2 + 9^2 + 2(100)(10) + 2(100)(9) + 2(10)(9) = 10000 + 100 + 81 + 2000 + 1800 + 180 =$ **14,161**.

Difference of Squares & Śhrīdharāchārya's Method (750 CE)

Difference of Squares Identity

$$a^2 - b^2 = (a + b)(a - b)$$

PRODUCT OF SUM AND DIFFERENCE

- **Rearranged Form:** $a^2 = (a + b)(a - b) + b^2$.
- **Geometrical Justification (Fig 4.5):** Take square a^2. Cut off corner b^2. Rearrange remaining L-strip into a rectangle of dimensions (a + b) by (a - b)!

Śhrīdharāchārya's Mental Squaring Trick

Proposed in **750 CE** by the great Indian mathematician Śhrīdharāchārya for lightning-fast mental squaring:

$55^2 = (55 + 5)(55 - 5) + 5^2 = 60 \times 50 + 25 = 3025$

$65^2 = (65 + 5)(65 - 5) + 5^2 = 70 \times 60 + 25 = 4225$

$85^2 = (85 + 5)(85 - 5) + 5^2 = 90 \times 80 + 25 = 7225$

The Vedic Rule for Numbers Ending in 5 (Ekādhikena Pūrevena)

For any number ending in 5, i.e., $(10n + 5)^2 = n(n + 1) \times 100 + 25$. Śhrīdharāchārya's identity explains the exact algebraic rationale behind this ancient shortcut!

Check Your Understanding: Trinomials & Fast Squaring

Q1 How does substituting $d = b + c$ enable you to prove the identity for $(a + b + c)^2$?

Q2 How many total terms appear in the expanded form of $(a + b + c)^2$, and how many are cross-products?

Q3 Using Śhrīdharāchārya's formula $a^2 = (a + b)(a - b) + b^2$, calculate 105^2 mentally by choosing $b = 5$.

Q4 In the geometric dissection of $a^2 - b^2$, what are the length and breadth of the rearranged rectangle?

Visualising (x + a)(x + b) with Algebra Tiles

Tiling Model for $x^2 + 7x + 12$

• Area of rectangle of sides **(x + 3) and (x + 4)**:

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

STANDARD MONIC QUADRATIC IDENTITY

- **1 square x^2-tile** of area x^2
- **7 linear x-tiles**: 3 placed to the right, 4 placed below
- **12 unit tiles** arranged in a 3 × 4 array at the corner
- Overall Dimensions formed: **(x + 3) by (x + 4)**!

General Linear Product: (px + a)(qx + b)

Consider dimensions **(2x + 3) and (3x + 1)** (Fig 4.8):

$$(px + a)(qx + b) = pq\,x^2 + (pb + qa)x + ab$$

GENERAL QUADRATIC TRINOMIAL PRODUCT

- 2 × 3 = **6 x^2-tiles**
- 2(1) + 3(3) = 2 + 9 = **11 x-tiles**
- 3 × 1 = **3 unit tiles**
- Total Area = **$6x^2 + 11x + 3$**.

Dual Action of Algebra Tiles

Tiles work in both directions: **Expansion** (multiplying linear dimensions to build a rectangle) and **Factorisation** (arranging a given set of tiles into a perfect rectangle to read off its factors!).

Middle-Term Splitting: Factoring Without Tiles

Case 1: $x^2 + 11x + 30$

- Compare with $x^2 + (a + b)x + ab$:
- Need: **$a + b = 11$** and **$ab = 30$**
- Test factor pairs of 30:
  — $2 \times 15 = 30$ (sum = 17 ❌)
  — $3 \times 10 = 30$ (sum = 13 ❌)
  — **$5 \times 6 = 30$ (sum = 11 ✅)**
- **Result:** $(x + 5)(x + 6)$

Case 2: $x^2 - 5x + 6$

- Middle coefficient is negative!
- Need: **$a + b = -5$** and **$ab = 6$**
- Since product is positive and sum is negative, both numbers must be negative:
  — **$(-2) \times (-3) = +6$**
  — **$(-2) + (-3) = -5$** ✅
- **Result:** $(x - 2)(x - 3)$

Case 3: $6x^2 + 7x + 2$

- Leading coefficient ≠ 1 (ac method):
- Product $ac = 6 \times 2 = \mathbf{12}$; Sum = **7**
- Numbers: **3 and 4** ($3 \times 4 = 12$, $3 + 4 = 7$)
- $6x^2 + 3x + 4x + 2$
  $= 3x(2x + 1) + 2(2x + 1)$
  $= \mathbf{(2x + 1)(3x + 2)}$

The Master Rule of Splitting

For $ax^2 + bx + c$, find two numbers p and q such that **$p + q = b$** and **$p \cdot q = a \cdot c$**. Grouping in pairs will then always expose a common binomial factor!

Check Your Understanding: Middle-Term Splitting

Q1 Why did splitting 7x as 2x + 5x fail to form a neat rectangle with 12 unit tiles in the algebra tile model?

Q2 To factor $s^2 - 11s + 24$, what two numbers must you find that multiply to 24 and add to -11?

Q3 How does the sign of the constant term tell you whether the two split numbers have the same sign or opposite signs?

Q4 In factoring $r^2 - r - 42$, what are the two integers whose product is -42 and sum is -1?

Cubes of Binomials: Geometric 3D Dissection

Expansion of $(a + b)^3$

• Multiply $(a + b)$ by $(a^2 + 2ab + b^2)$:

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

THE BINOMIAL CUBE IDENTITY

• **3D Geometric Dissection (8 Pieces):**

— **2 Cubes:** a^3 and b^3

— **3 Cuboids:** dimensions $a \times a \times b$ (volume = $\mathbf{3a^2b}$)

— **3 Cuboids:** dimensions $a \times b \times b$ (volume = $\mathbf{3ab^2}$)

Expansion of $(a - b)^3$

• Substitute **b** → **−b** in $(a + b)^3$:

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

ALTERNATING SIGN PATTERN

• **Key Observation:** The signs alternate strictly: (+, −, +, −).

• **Factoring Example 14:**

$8n^3 - 60n^2m + 150nm^2 - 125m^3$

$= (2n)^3 - 3(2n)^2(5m) + 3(2n)(5m)^2 - (5m)^3$

$= \mathbf{(2n - 5m)^3}$.

Check Your Understanding: Binomial Cubes

Q1 Into how many individual 3-dimensional blocks (cubes and cuboids) is a cube of edge (a + b) partitioned?

Q2 Why does the term $3ab^2$ carry a positive sign in the expansion of $(a - b)^3$, while $-3a^2b$ carries a negative sign?

Q3 If a solid cube has a volume given by $p^3 + 6p^2q + 12pq^2 + 8q^3$, what is the exact length of its edge?

Q4 How can you use the identity for $(a - b)^3$ to evaluate 99^3 without performing standard long multiplication?

Sum & Difference of Cubes: $x^3 - y^3$ and $x^3 + y^3$

− Difference of Cubes

• Multiply $(x - y)$ by $(x^2 + xy + y^2)$:

$= x(x^2 + xy + y^2) - y(x^2 + xy + y^2)$

$= (x^3 + x^2y + xy^2) - (x^2y + xy^2 + y^3)$

$= \mathbf{x^3 - y^3}$

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

LINEAR FACTOR: (X − Y)

+ Sum of Cubes

• Multiply $(x + y)$ by $(x^2 - xy + y^2)$:

$= x(x^2 - xy + y^2) + y(x^2 - xy + y^2)$

$= (x^3 - x^2y + xy^2) + (x^2y - xy^2 + y^3)$

$= \mathbf{x^3 + y^3}$

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

LINEAR FACTOR: (X + Y)

Think & Reflect: The Universal (x − y) Factor

Notice that $(x - y)$ is a factor of $x^2 - y^2$ and $x^3 - y^3$. Indeed, for any positive integer n, **$(x - y)$ is ALWAYS a factor of $(x^n - y^n)$**! For example, $x^4 - y^4 = (x^2 - y^2)(x^2 + y^2) = (x - y)(x + y)(x^2 + y^2)$.

The Three-Variable Cubic Identity: $x^3 + y^3 + z^3 - 3xyz$

The Master 3-Variable Identity

$$(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) = x^3 + y^3 + z^3 - 3xyz$$

Worked Example 15

Given: $x + y + z = 10$, $xyz = 25$, $x^2 + y^2 + z^2 = 38$.

To find: $x^3 + y^3 + z^3$

1. From $(x + y + z)^2 = (x^2 + y^2 + z^2) + 2(xy + yz + zx)$:
$100 = 38 + 2(xy + yz + zx) \Longrightarrow$ **$xy + yz + zx = 31$**

2. Substitute into identity:
$(10)(38 - 31) = x^3 + y^3 + z^3 - 3(25)$
$70 = x^3 + y^3 + z^3 - 75 \Longrightarrow$ **$x^3 + y^3 + z^3 = 145$** ✅

The Conditional Corollary

What happens if **$x + y + z = 0$**?

$$\text{If } x + y + z = 0 \Longrightarrow x^3 + y^3 + z^3 = 3xyz$$

THE ZERO-SUM CUBIC THEOREM

Allows direct computation of expressions like $28^3 + (-15)^3 + (-13)^3 = 3(28)(-15)(-13)$ without cubing!

Check Your Understanding: Cubic Identities

Q1 What is the middle term in the trinomial factor of the sum of cubes: $x^3 + y^3$?

Q2 If $a + b + c = 0$, what simple value does the expression $a^3 + b^3 + c^3$ reduce to?

Q3 Why does expanding $(x - y)(x^2 + xy + y^2)$ cause all intermediate terms like x^2y and xy^2 to cancel out?

Q4 Show how $(x - y)$ can be proven to be an algebraic factor of $x^4 - y^4$.

Simplifying Rational Algebraic Expressions

Method of Factor & Cancel (Ex 16)

Problem: Simplify $(x^2 - 7x + 12) / (5x^2 + 5x - 100)$.

- **Numerator:** $x^2 - 7x + 12 = \mathbf{(x - 3)(x - 4)}$
- **Denominator:** $5(x^2 + x - 20) = \mathbf{5(x - 4)(x + 5)}$
- Cancel non-zero common factor $(x - 4)$:

$$(x - 3) / [5(x + 5)]$$

SIMPLIFIED RATIONAL FORM

Real-World Geometry: The Pool (Ex 18)

- Rectangular pool breadth = $(x - 4)$ m, length = x m, Area = 96 m².
- Equation: $x(x - 4) = 96 \implies x^2 - 4x - 96 = 0$
- Split $-4x$ into $-12x + 8x$:
  $(x - 12)(x + 8) = 0 \implies x = 12$ or $x = -8$
- Length cannot be negative! So **Length = 12 m**, **Breadth = 8 m**.

Critical Precondition for Cancelling

A factor $(x - k)$ can be cancelled from numerator and denominator **if and only if** it is strictly non-zero (i.e. $x \neq k$). Division by zero is mathematically undefined!

⚡ QUICK BOX 8 • CLASSROOM ORAL CHECK

Check Your Understanding: Rational Simplification

Q1 Why must we explicitly state that the denominator is non-zero before cancelling factors in a rational expression?

Q2 If Saira forms a rectangle with tiles giving area $x^2 + 8x + 15$, what are its dimensions in terms of x?

Q3 In the swimming pool problem $x^2 - 4x - 96 = 0$, why did we discard the algebraic root $x = -8$?

Q4 What common factor cancels out when simplifying $(4x^2 + 4x + 1) / (4x^2 - 1)$?

Representative Worked Problems from the Chapter

Equal Coefficients (Ex 10)

Problem: If $(x - 2)$ and $(x - \frac{1}{2})$ are factors of $px^2 + 5x + r$, show $p = r$.

Solution:

- Root $x = 2 \Longrightarrow 4p + 10 + r = 0$ (1)
- Root $x = \frac{1}{2} \Longrightarrow p/4 + 5/2 + r = 0$

$\Longrightarrow p + 10 + 4r = 0$ (2)

- Subtract (2) from (1):

$3p - 3r = 0 \Longrightarrow \mathbf{p = r}$

Proved: p = r ✅

Cubic Evaluation (Ex 11)

Problem: $a + b + c = 5$, $ab + bc + ca = 10$. Prove $a^3 + b^3 + c^3 - 3abc = -25$.

Solution:

$a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$

$= 5^2 - 2(10) = \mathbf{5}$.

Identity $= (a+b+c)[(a^2+b^2+c^2) - (ab+bc+ca)]$

$= 5 \times (5 - 10) = 5 \times (-5) = \mathbf{-25}$

Proved: Value is −25 ✅

Zero Substitution (Ex 13)

Problem: Value of $x^3 + y^3 - 12xy + 64$ when $x + y = -4$.

Solution:

Let $z = 4$. Then $x + y + z = x + y + 4 = 0$!

When sum $= 0$, $x^3 + y^3 + z^3 = 3xyz$.

$x^3 + y^3 + 4^3 = 3(x)(y)(4) = 12xy$

$\Longrightarrow x^3 + y^3 - 12xy + 64 = \mathbf{0}$

Ans: Exactly 0 ✅

CLASSROOM DISCUSSION • ADVANCED INQUIRIES

Pages 89-90 (Problems 7, 8, 12)

Extended Discussions & Number Theory Inquiries

Divisibility of $n^3 - n$ (Problem 12)

Factor the expression:
$n^3 - n = n(n^2 - 1) = (n - 1) \cdot n \cdot (n + 1)$

Teacher Discussion Prompt: Notice this is the product of **3 consecutive integers**! In any 3 consecutive numbers, at least one is divisible by 2 and exactly one is divisible by 3. Why does this guarantee divisibility by $2 \times 3 = 6$?

Number + Reciprocal (Problem 8)

A number x satisfies **$x + 1/x = 10/3$**.

- Multiply by 3x: $3x^2 - 10x + 3 = 0$
- $(3x - 1)(x - 3) = 0$
- **Roots:** $x = 3$ or $x = 1/3$

Teacher Discussion Prompt: Why do the two solutions naturally turn out to be mutual reciprocals of one another?

Playground Path Area (Problem 7)

A square ground of side 40 m surrounded by a walkway of uniform width s metres.

- Outer square side = $(40 + 2s)$
- Path Area = $(40 + 2s)^2 - 40^2$
$= (40 + 2s - 40)(40 + 2s + 40)$
$= 2s(80 + 2s) =$ **$160s + 4s^2$**

Teacher Prompt: How does difference of squares make this calculation effortless?

Complete Concept Checklist & Core Takeaways

Degree 2 Identities & Factoring

- ✅ **$(x + y)^2 = x^2 + 2xy + y^2$:** Area of square of side (x + y).
- ✅ **$(x - y)^2 = x^2 - 2xy + y^2$:** Square of binomial difference.
- ✅ **$x^2 - y^2 = (x + y)(x - y)$:** Difference of squares (Śhrīdharāchārya).
- ✅ **$(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$:** Trinomial square.
- ✅ **$(x + a)(x + b) = x^2 + (a + b)x + ab$:** Monic quadratic product.
- ✅ **$(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd$:** General linear product.

Degree 3 Identities & Rational Simplification

- ✅ **$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$:** Dissection into 2 cubes + 6 cuboids.
- ✅ **$(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$:** Alternating sign cubic expansion.
- ✅ **$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$:** Factoring difference of cubes.
- ✅ **$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$:** Factoring sum of cubes.
- ✅ **$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$:** 3-variable cubic.
- ✅ **Rational Expressions:** Simplify by cancelling common non-zero factors.

END OF CHAPTER 4

Identities are the Universal Keys That Unlock the Patterns of Algebra.

"In every geometric dissection lives an algebraic truth, and in every identity lies the power to simplify the complex."

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